

Cyber Security Internship

Design a Cyber Incident Response Plan

MITRE ATT&CK Framework

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Organization: Internee.pk

➤ Objective

Develop a structured response plan for handling cybersecurity incidents effectively.

➤ Incident Response Plan

An Incident Response Plan (IRP) is a structured process that helps an organization detect, respond to, and recover from cybersecurity incidents effectively. It minimizes the impact of security breaches, protects organizational assets, and ensures business continuity. This report presents a simulated ransomware attack scenario and demonstrates the steps taken to detect, contain, eradicate, and recover from the incident following industry best practices.

Simulated Ransomware Scenario

A staff member at **Internee.pk** receives a phishing email with an attachment named "**Internship_Applications_2026.pdf.exe**". Believing it to be a legitimate document, the employee opens the attachment, which executes ransomware on the organization's Ubuntu web server.

Within a few minutes, important files become inaccessible, the website starts malfunctioning, and a ransom note appears demanding payment in cryptocurrency to recover the encrypted files. The organization's security team immediately activates the Incident Response Plan to minimize damage and restore normal operations.

➤ Incident Response Lifecycle

Phase 1 – Preparation

Before the incident occurred, the organization implemented several security measures, including regular data backups, antivirus protection, firewall configuration, multi-factor authentication, and employee cybersecurity awareness training. An incident response team was assigned clear responsibilities for handling cybersecurity incidents.

Phase 2 – Detection and Analysis

The attack was identified after employees reported that important files could no longer be accessed. Security logs showed multiple suspicious activities, and unusual CPU usage was observed on the server. A ransom note appeared on the infected system, confirming the presence of ransomware. The security team analyzed system logs and verified the compromise.

Phase 3 – Containment

To prevent the ransomware from spreading, the infected server was immediately isolated from the network. Remote access to the compromised system was disabled, suspicious user accounts were temporarily locked, and network communication with potentially malicious IP addresses was blocked. System logs and evidence were preserved for further investigation.

Phase 4 – Eradication

The security team removed the malicious files from the infected system and performed a complete malware scan. Security patches were applied to vulnerable services, passwords for affected accounts were reset, and the server was verified to ensure that no malicious processes remained active.

Phase 5 – Recovery

Clean backup files were restored to the server, and all critical services were tested to verify proper functionality. The web server was gradually returned to production while continuous monitoring was performed to ensure that no further malicious activity occurred after recovery.

Phase 6 – Lessons Learned

After resolving the incident, the organization reviewed the response process and identified areas for improvement. Additional phishing awareness training was planned for employees, backup procedures were reviewed, and monitoring capabilities were strengthened. The Incident Response Plan was updated based on the findings from the simulated attack.

➤ Staff Awareness

Employees play an important role in preventing ransomware attacks. Staff members should avoid opening suspicious email attachments, verify unknown senders before downloading files, immediately report suspicious emails to the IT department, use strong passwords, and enable multi-factor authentication whenever possible.

➤ MITRE ATT&CK Mapping

Attack Stage	MITRE Technique
Phishing Email	T1566 – Phishing
User Execution	T1204 – User Execution
Data Encryption	T1486 – Data Encrypted for Impact

➤ Tools Used

Tool	Purpose
MITRE ATT&CK Framework	Used to map attacker techniques during the simulated attack.
System Logs	Used to identify suspicious activities and security events.
Microsoft Defender (Simulation)	Used to detect malicious activity during the simulation.
VirusTotal (Simulation)	Used to analyze the suspected ransomware sample.

➤ **Verification**

The effectiveness of the incident response process was verified by confirming that the ransomware was detected successfully, the infected system was isolated, malicious files were removed, clean backups were restored, and all critical services returned to normal operation. Security logs were reviewed to ensure that no further suspicious activities were detected after recovery.

➤ **Conclusion**

This simulated ransomware incident demonstrated the importance of having a well-defined Cyber Incident Response Plan. By following the phases of preparation, detection, containment, eradication, recovery, and lessons learned, the organization was able to minimize the impact of the attack and restore normal operations efficiently. The simulation also emphasized the value of employee awareness, regular backups, and continuous monitoring in strengthening an organization's cybersecurity posture.